

Book Shop

Time limit: 1.00 s

Memory limit: 512 MB

You are in a book shop which sells n different books. You know the price and number of pages of each book. You have decided that the total price of your purchases will be at most x . What is the maximum number of pages you can buy? You can buy each book at most once.

Input

The first input line contains two integers n and x : the number of books and the maximum total price.

The next line contains n integers $h_1, h_2, \dots, h_n$: the price of each book.

The last line contains n integers $s_1, s_2, \dots, s_n$: the number of pages of each book.

Output

Print one integer: the maximum number of pages.

Constraints

- $1 \leq n \leq 1000$
- $1 \leq x \leq 10^5$
- $1 \leq h_i, s_i \leq 1000$

Example

Input	Output
4 10 4 8 5 3 5 12 8 1	13

Explanation: You can buy books 1 and 3. Their price is $4 + 5 = 9$ and the number of pages is $5 + 8 = 13$.